CLIMATE ACTION RESERVE

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Request for Project Variance

The Climate Action Reserve (Reserve) will allow variances in the way specific projects monitor or calculate greenhouse gas emission reductions only where Reserve staff determine that such variances are acceptable and upon submission of this form and payment of the $1,350 fee. No variance request will be considered until the project in question has been formally submitted to the Reserve and all invoices have been paid.

Only with explicit, written acceptance of the variance may a project developer apply monitoring or calculation methods not contained in the Reserve's applicable protocol for their project. In some cases, a variance request may be granted only for a specified time period. Decisions on variances are to be considered *sui generis*, and are not precedent-setting. Variances will not be granted for exceptions to project eligibility criteria or to the Reserve's methodological approaches to determining GHG emission reductions. If the requested variance diverges significantly from the approved methodology in that it requires extensive analysis of site specific features and/or employs concepts not fully vetted through public consultation, the variance must be denied. Only one issue (related to a specific rule or requirement in the protocol) may be submitted per variance request. If multiple variances are needed, please submit a separate variance request per issue. Please provide complete information at the time of the request, as re-opening or adding to an existing determination will be considered a new variance request. Each new or re-opened variance request will incur the $1,350 fee.

The Reserve retains the right to reject outright any Request for Project Variance, request further documentation, or impose additional constraints and/or discount factors on the proposed monitoring or calculation methods.

If the Reserve accepts this variance request, a letter describing the variance granted will be sent to the project developer and will be made publicly available. Variance determinations are final.

Instructions:

1. Upload the Request for Project Variance to the *Project Documents* section of the project for which the project variance request applies. Please provide as much detail and supporting documentation as possible with your request to facilitate our review.
2. An invoice for the Variance Review Fee ($1,350.00 per request) will be posted in the *My Invoices* section of your account.
3. Pay the Variance Review Fee (instructions for payment are located on the invoice).
4. The Request for Project Variance review will commence upon completion of steps 1-3. At this point Reserve staff may contact you for additional information – timing of the issuance of a variance determination will depend on the completeness of information provided.

Project Details

Date of Request: **5/25/2023**

Project Developer: **Indigo Ag.**

Project Name: **Indigo U.S. Project No. 1 (CAR1459)**

Project Type: **Soil Enrichment**

Protocol Version Used: **SEP Version 1.1**

Reporting Period (for which request applies to): **RP3 (ending 12/31/2022)**

Form Completed By (name and email): **Ryan Pape rpape@indigoag.com**

Requested Variance

1. Protocol rule/requirement in question (include page number and quoted text):

This request includes variance to section 5, page 29 of SEP version 1.1, which reads in part:

" GHG emission reductions from a soil enrichment project are quantified by comparing modeled and calculated project and baseline emissions, as well as from calculating changes in SOC."

As well as section 6, page 67 of SEP Version 1.1 which reads in part.

"9. Modeling plan, if applicable a. The project monitoring plan will identify the model(s) selected initially and document analysis and results demonstrating validation of the model(s). Model validation datasets will be archived to permit periodic application to calculate model structural uncertainty. The modeling plan will detail all required model input parameters and specify the baseline schedule of agricultural management activities for each sample unit"

2. Proposed variance:

This variance, described in detail in the attached, seeks to utilize a conservative approach to estimate grazing impacts on CO_2 emissions from the soil carbon pool, using a combination of approved DayCent-CR modeling, SEP derived default equations, and literature-based estimates. N_2O and CH_4 emissions will be calculated using the relevant SEP default equations.

3. Reason why the project does not meet the rule/requirement listed above:

Currently there are not sufficient published datasets for grazing impacts on soil carbon in cropland systems available to meet the model calibration and validation requirements for DayCent-CR. This variance serves as a conservative, short-term bridge to qualify the fields in question while additional data is collected to satisfy cal/val requirements for future reporting periods.

4. Conservative, compensatory measures (e.g., discount factors) taken to counteract variance:
    - **A literature survey shows that the $Frac_{stabilization}$ factor chosen for estimation calculations is well on the**

conservative end of the spectrum of values that have been accepted in literature.

- **The incorporation of manure C into SOC is widely believed to be more efficient than equivalent amounts of C derived from crop residue, this approach conservatively assumes an equal stabilization efficiency.**

- **A conservative digestibility factor (DE) was chosen, using the mean of the values table A-139. (EPA, 2023) This factor is considered more conservative than values we would expect in the field, as more of the grazing events relevant to this variance come from stover with a lower digestibility factor and thus a higher likelihood of returning more C to the soil. Most of the crops in Table A-139 are grass species with higher digestibility, so the average will be conservative.**

- **Cropping experiments also often report yield increases from improved nutrient cycling on plots with grazing activities; these increases in productivity are conservatively not included in these calculations.**

- **Finally, two recent studies find that livestock integration to croplands positively impacts a broad range of soil health indicators. Conservatively, these positive factors are not included in these calculations, only a negative emission factor will be possible from this calculation.**

5. Justification for why variance results in equal or greater conservativeness than protocol:

Conservative $Frac_{stabilization}$ and DE factors, in combination with the exclusion of several factors which are widely accepted on their own to increase SOC or improve general soil and crop health, suggest that the variance approach is more conservative than what a complete modeling solution approved by SEP V1.1 is likely to provide, should version 2 of the SEP incorporate different guidance for how SOC impacts from grazed cropland may be estimated outside of a biogeochemical model, the

quantification from any reporting period conducted using the guidance in this variance will be reviewed to ensure compliance with the language in V2.0. If the updated guidance would result in fewer CRTs from those reporting periods a commensurate adjustment will be made to the quantification during the next verification.

6. Additional details or documentation to facilitate our review of the variance request: (attach additional files as needed, such as flow data, calibration certificates, etc.)

Please refer to the complete proposal on the following pages.

Proposal to estimate grazing impacts on CO_2 emissions using a simple, conservative approach to allow crediting of grazed fields

Proposed by: Indigo Ag
Submitted: May 25th, 2023

1. Description of Proposal

The Soil Enrichment Protocol requires formal model validation for all practices included in a Project's modeling domain. Emissions of N_2O and CH_4 associated with grazing activities can be quantified using SEP default equations (SEP Eqns 5.11b, 5.13b, 5.22 & 5.23), but emissions and removals of CO_2 associated with grazing impacts on soil organic carbon (SOC) stocks must be quantified by a validated model.

Many fields in Indigo's project CAR 1459 are enrolled based on practice changes (PCs) for which Indigo's biogeochemical model (DayCent-CR) is validated under the SEP, but also include grazing events in their historical baseline and/or project management data. Grazing is currently outside of the domain of the approved model cal/val report for DayCent-CR. Thus, Indigo has so far not included any fields within grazing in the quantification and verification of CAR1459. However, we have been making payments to these fields in order to keep them in the program until we can quantify their impacts.

This proposal presents an approach to quantifying emissions from such fields without a model validated for grazing, consisting of four parts (see Figure 1):

1. Quantify all sources of N_2O and CH_4 associated with grazing activities using SEP default equations.
2. Model changes in SOC stocks from all agronomic activities, excluding grazing activities, to quantify net soil CO_2 emissions. (i.e., current modeling approach for CAR1459).
3. Quantify the impact of grazing activities on SOC stocks using the simple, conservative approach given in Equation 1, below.
4. Subtract this estimated impact from the modeled SOC impacts.

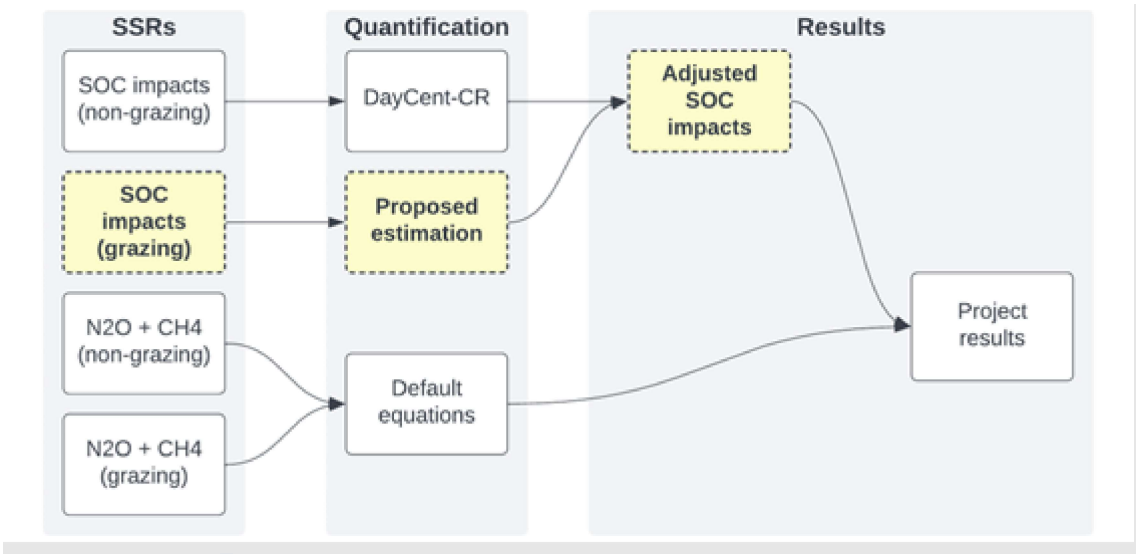


Figure 1 Outline of proposed approach for including non-validated grazing in CAR1459

2. Proposed Quantification of Grazing Impacts on SOC

While there may be complex soil dynamics from grazing that, in certain areas, prove beneficial to the SOC stocks, the main impact from grazing will be a reduction in carbon inputs to the soil due to the removal of plant biomass as forage. The proposed approach accounts for all emissions from grazed fields using SEP-specified methods except for the conversion of a fraction of forage biomass C into CO_2, which reduces the amount of C input to SOC stocks. Equation 1 proposes to account for this "C opportunity cost" by using parameters from standard references to estimate the amount of C respired by grazing animals, and using a literature-based estimate of the fraction of aboveground forage biomass incorporated into SOC.

Equation 1. Carbon dioxide losses due to grazing activities not included in SOC modeling

$$CO2_{grazing_SOC} = \Sigma\ (AGD_l \times TAM_l \times DMI \times Frac_C \times DE_f \times Frac_{stabilization} \times 44/12 \times 1/1000$$

Table 1. Proposed parameters from Equation 1

Parameter	Proposed Definition	Data Unit	References
$CO2_{grazing_SOC}$	forage biomass C emissions from grazing livestock respiration that would otherwise have been incorporated into stabilized SOC	tCO2e/acre	
AGD_l	Animal grazing days for livestock category *l*. Per SEP Section 5.5.1, the minimum value allowed for the project scenario is equal to the average value from the historical baseline period	animal days/acre	CAR SEP
TAM_l	Typical animal mass for livestock category *l*; animal type mappings are shown in Table 3	kg/animal	(EPA, 2023), Table A-158
DMI	Dry matter intake by grazing animals per animal mass, per day; fixed at .03	kg DM/kg animal mass/day	(Dong et al., 2006), page 10.23
$Frac_C$	C content of forage dry matter; fixed at 0.40	kg C/kg dry matter	
DE	Digestibility of grazed forage; fixed at 0.655 based on average value across all forage types	kg C respired/kg dry matter intake	(EPA, 2023), table A-139
$Frac_{stabilization}$	Fraction of forage biomass C that is incorporated into stable SOC stocks; estimated at 0.25	kg SOC-C/kg biomass C	See Rationale and References
44/12	Molar mass ratio to convert C to CO_2e	kg CO_2e/kg SOC-C	
1/1000	kg to tons unit conversion	tons/kg	

3. Rationale for proposed approach

The proposed approach quantifies grazing impacts on SOC by first estimating the amount of forage C respired by grazing livestock, and then estimating the portion of that respired C that would otherwise have been stabilized into stable SOC pools. This set of parameters and mass-balance logic are essentially the same as those that would be used to generate inputs for biogeochemical modeling of these activities, except for the $Frac_{stabilization}$ parameter. This parameter depends on several factors including residue chemistry, site management and pedoclimatic factors. We surveyed the peer-reviewed literature and present several measurement-derived estimates of $Frac_{stabilization}$ in Table 2. Given the wide range of estimates for this parameter, we chose a conservative value of 0.25.

Table 2. Literature estimates for efficiency of residue C stabilization into SOC

Source	Description	$Frac_{stabilization}$ Estimate
(Kätterer et al., 2011)	Estimates of humification coefficients for several types of organic materials based on long-term Swedish cropping experiment; shown is estimate for shoot-derived C	0.17
(Poeplau et al., 2015)	Modeled estimate of humification coefficient for aboveground crop litter based on several European cropping experiments	0.0-0.03
(Allmaras et al., 2004)	Estimates of crop-derived SOC based on natural abundance ^{13}C measurements from a Minnesota long-term experiment	No-till: 0.26 Conventional till: 0.11
(Collins et al., 1999)	Estimates of % corn residue remaining in SOC based on measurements of natural abundance ^{13}C at eight long-term cropping experiments in the US Midwest	0.055-0.16
(Kong & Six, 2010)	Stable isotope labeling experiment in California to measure incorporation of aboveground vs belowground crop C into SOC; shown is aboveground estimate	0.04

The choice of 0.655 for a digestibility estimate is conservative because it reflects an average that includes many highly-digestible, fast-growing forage types. In reality, we believe most of these grazing activities were on dead crop residues (e.g., corn stover) with relatively low digestibility, which would imply a larger fraction of consumed biomass C being returned in feces than the chosen parameter value assumes.

Several additional factors suggest that the grazing activities on these fields are conservatively quantified by this approach. First, the incorporation of manure C into SOC is widely believed to be more efficient than equivalent amounts of crop-residue C for complex chemical and biological reasons (Kätterer et al., 2011; Prairie et al., 2023; Tracy & Zhang, 2008), whereas this approach conservatively assumes equal stabilization efficiencies. Second, cropping experiments often report yield increases from improved nutrient cycling on plots with grazing activities compared to conventionally managed controls (Drewnoski & Blanco, 2015; Rasby et al., 2014; Tracy & Zhang, 2008). Finally, two recent studies find that livestock integration to croplands positively impacts a broad range of soil health indicators, including particulate organic matter, aggregation, and microbial diversity (Bansal et al., 2022; Prairie et al., 2023).

4. References

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5. Appendix: Parameter Data Tables

Table 3. Proposed values for Typical Animal Mass (TAM_i) per animal type, from EPA 2023 Table A-158

SEP Animal Type	EPA 2023 Animal Group	Typical Animal Mass (kg)
Dairy Cows	Dairy Cows	680
Dairy Heifers	Dairy Heifers	408
Bulls	NOF Bulls	917
Calves	NOF Calves	118
Beef Heifers	NOF Heifers	407
Steers	NOF Steers	335
Beef Cows	NOF Cows	611
Bison	American Bison	578.5
Swine (average for all weight categories)	Market Swine >180 lbs.	91
Sheep	NOF Sheep	80
Goats	Goats	64
Horses	Horses	450
Mules and Asses	Mules and Asses	130
Poultry (using values from turkeys)	Turkeys	6.8